

TREVOR CHRISTENSEN

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RESEARCH SUMMARY

PhD student developing machine-learning approaches for genomic sequence-to-function prediction. My research focuses on scaling laws, active learning, knowledge distillation, and uncertainty quantification that make models more accurate, efficient, and cost-effective to train and deploy. I am motivated by research with real-world applications that improve human health and wellbeing.

EDUCATION

Cold Spring Harbor Laboratory — Ph.D. in Biological Sciences **August 2024 – present**
T32 training grant recipient • George A. and Marjorie H. Anderson Fellow • Cold Spring Harbor, NY

Claremont McKenna College — B.A. in Molecular Biology **August 2018 – May 2022**
GPA 3.99/4.00 • summa cum laude • Dean's List (every year) • Claremont, CA

CSHL–Hofstra Bioscience Business Innovation Program **September 2025 – March 2026**
Training in translating scientific research into commercial and clinical strategy.

RESEARCH EXPERIENCE

Cold Spring Harbor Laboratory — Doctoral Researcher, Koo Lab **May 2025 – present**
Advisor: Peter K. Koo (joined after first year of coursework and rotations), Cold Spring Harbor, NY

- Developing data scaling strategies for genomic sequence-to-function models using active learning and other strategies that identify which sequences to generate to maximize model performance per experiment and per dollar, informing the design of large shared perturbation datasets
- Ran knowledge-distillation experiments for genomic regulatory-activity prediction, producing models with $\sim 10\times$ reduced inference time at comparable performance, deployable across clinical, academic, and industrial settings (co-author, npj Artificial Intelligence, 2026).

National Cancer Institute, Bethesda, MD — Postbaccalaureate Fellow **May 2022 – June 2024**
Advisors: Jason W. Hoskins, Ph.D., and Laufey T. Amundadottir, Ph.D., Rockville, MD

- Characterized GWAS SNPs associated with pancreatic cancer risk, using base editing and multi-omics approaches to discover biomarkers and therapeutic targets (second-author medRxiv preprint, 2026)
- Developed a computational protocol to infer gene-expression regulators from RNA-seq and eQTL data, revealing how pathogenic variants rewire regulatory networks (second-author STAR Protocols, 2023)

Keck Science Department — Research Assistant, Leconte Lab **January 2020 – May 2022**
Advisor: Aaron M. Leconte, Ph.D., Claremont, CA

- Engineered DNA polymerases to synthesize 2'-fluoro-modified nucleic acids with $2\times$ higher accuracy and longer synthesis length, with applications to synthetic biology, aptamers, and other nucleic-acid therapeutics (first-author RSC Chemical Biology, 2022).
- Designed directed evolution screens to further improve enzyme properties

SKILLS & PROFICIENCIES

Programming: Python (PyTorch, TensorFlow), R, Git, high-performance computing

Machine learning: Sequence-to-function models, deep learning, knowledge distillation, uncertainty quantification, active learning, ensemble methods, transfer learning, attribution/interpretability analysis

Tools and Methods: RNA-seq analysis, gene-regulatory-network inference, variant effect prediction, GWAS, CRISPR/Cas9 base editing, modified-nucleotide synthesis, nucleic-acid biochemistry

Wet lab: PCR, cell culture, transfection, gel electrophoresis, imaging, DNA/RNA extraction & sequencing, experimental design

PUBLICATIONS

Zhou, J., Rizzo, K., **Christensen, T.**, Tang, Z., & Koo, P. K. (2026). Uncertainty-aware genomic deep learning with knowledge distillation. *npj Artificial Intelligence*, 2, 3. <https://doi.org/10.1038/s44387-025-00053-3>

Hoskins, J. W., **Christensen, T. A.**, Eiser, D., Char, E., Mobaraki, M., O'Brien, A., Collins, I., Zhong, J., Patel, M. B., Prasad, G., Pancreatic Cancer Cohort Consortium and Pancreatic Cancer Case-Control Consortium (PanScan/PanC4), Arda, H. E., Connelly, K. E., & Amundadottir, L. T. (2026). Characterization of a pancreatic cancer GWAS signal suggests PDX1 buffers stress in the exocrine pancreas. *medRxiv* (preprint). <https://doi.org/10.64898/2026.04.13.26350790>

Hoskins, J. W., **Christensen, T. A.**, & Amundadottir, L. T. (2023). Master regulator activity QTL protocol to implicate regulatory pathways potentially mediating GWAS signals using eQTL data. *STAR Protocols*, 4(3), 102362. <https://doi.org/10.1016/j.xpro.2023.102362>

Christensen, T. A., Lee, K. Y., Gottlieb, S. Z. P., Carrier, M. B., & Leconte, A. M. (2022). Mutant polymerases capable of 2' fluoro-modified nucleic acid synthesis and amplification with improved accuracy. *RSC Chemical Biology*, 3(8), 1044–1051. <https://doi.org/10.1039/d2cb00064d>

PRESENTATIONS

CSHL Symposium: AI in Biology, Cold Spring Harbor, NY **May 2026**
A Data-Centric Approach to Improving Sequence-Function models in Regulatory Genomics

Stowers Research Conference, Kansas City, MO **April 2026**
A Data-Centric Approach to Improving Sequence-Function models in Regulatory Genomics

Postbaccalaureate Poster Day, Bethesda, MD **April 2023, 2024**
Functional Characterization of Pancreatic Cancer GWAS Signal at PDX1 Promoter

- Won Outstanding Poster Award (given to top 20% of posters by a panel of judges)

American Society of Human Genetics 2023 Meeting, Washington, DC **November 2023**
Functional Characterization of Pancreatic Cancer GWAS Signal at PDX1 Promoter

American Chemical Society Spring 2022 Symposium, San Diego, CA **March 2022**
Improving M-DNA Polymerase Accuracy Through Rational Design

TEACHING & SERVICE

DNA Learning Center — Course Instructor **January 2025 – June 2025**

- Taught molecular biology and genetics to middle- and high-school students, leading hands-on labs in PCR and DNA extraction.

Program for Advancing the Health Sciences — Volunteer Mentor **August 2022 – June 2023**

- Mentored high-school students through independent research projects and health-science career planning.

Keck Science Department — Teaching Assistant & Peer Mentor **September 2019 – May 2022**

- Led wet- and dry-lab activities for undergraduate chemistry and biology courses and trained new lab members in experimental techniques.

REFERENCES

Available upon request